University of Central Punjab
Faculty of Information Technology

PROGRAM (S) TO BE EVALUATED

BSCS, PADP

A. Course Description

Course Code	AR332 (Theory) + AR331 (Lab)
Course Title	Operating Systems
Credit Hours	3 (2+1)
Prerequisites by Course(s) and Topics	Data Structures and Algorithms
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	• **Theory (3 credit hr.)** Final Exam 40% Mid Term exam 20% **Assignments (3)** 15% Quizzes (3) 20% Class Participation 05% **Theory Total** **100%** • **Lab (1 credit hr.)** Final Lab Exam 40% Lab Midterm 20% Graded Labs 15% Class participation 05% Project 20% **Lab Total** **100%**
Course Instructors	
Course Coordinator	**Hafiz Bilal Shahid**
Lab Instructors	
Plagiarism Policy	Cheating is not only unethical; it is also against the University rules and will not be tolerated. Any form of cheating in programming assignments, homework problems, quizzes, and exams will result in strict action. Plagiarism detection tools will be used to determine who has cheated in programming assignments. **All the parties involved will be awarded Zero in first instance. Second offense will result in three instruments being marked zero and on third offense case be forwarded to Disciplinary committee for an F grade in course.**
Attendance Policy	Be in class on time, attendance will be marked and closed in the first five minutes of the class.
Submission Policy for assignments	Manual assignments: 1. All manual assignments must be handwritten on a clear A4 page 2. No file or lamination should be done 3. Marks will be awarded for neatness Coding assignments: 1. All assignments are to be submitted on Teams/Portal. 2. Email submissions will not be considered under any circumstance
Late submission Policy	**24 hours late with 25% deduction in marks** **48 hours late with 50% deduction in marks** **More than 48 hours, 100% deduction**

Current Catalog Description	This is designed to be the first undergraduate level course with the main objective of teaching students the concepts and principles that underlie the design and implementation of contemporary operating systems. The discussion of the concepts and principles used in UNIX, Linux, and MS-DOS/MS-Windows operating systems will be integrated throughout the course. Students will be given the opportunity to examine important UNIX and Linux data structures and use its various services through programming exercises. Programming language environment: C on UNIX/Linux			
Grading	All the grading will be posted on portal within a week of evaluation. Numbers posted will be deemed final after 1 week of posting(i.e. can be contest within 7 days)			
Textbook (or **Laboratory Manual** for Laboratory Courses)	• Silberschatz, Galvin and Gagne, Operating System Concepts, 9th Edition. • Operating Systems Internals and Design Principles by Willam Stallings 5th edition			
Reference Material	● Tanenbaum, Andrew S., Modern Operating Systems, 3rd Edition, Prentice Hall. ● E-book is available at https://bit.ly/3VJVIhv			
Course Goals	● Students have developed a thorough understanding of the principles and concepts involved in the design of operating systems. ● Students have an in-depth understanding of the various OS services for threads, inter-process communication, process synchronization, and file systems, offered as system or library calls in UNIX and Linux operating systems.			
Course Learning Outcomes				
Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)	Attached			
Programming Assignments Done in the Course	Yes			
Class Time Spent on (in credit hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	1.5	1.0	1.0	0.5
Oral and Written Communications				

CLO#	Course Learning Outcome (CLO)	Taxonomy Level	Mapping to PLO
CLO 1	**Explain** basic structure, characteristics and core functionalities of Operating Systems and their components.	C2	2
CLO 2	**Demonstrate** knowledge of software solutions to various OS problems that have been developed for provisioning of required functionality and services in operating systems.	C3	2
CLO 3	**Compare** algorithms and techniques related to various OS services and core functionality, along with their performance issues.	C4	3

WEEKLY BREAKDOWN

Week #	Topics Covered	Book referemce	Evaluation Used	Relation to CLO
1	**Introduction to the Course** ▪ What is an Operating System? ▪ Why do we need an OS? ▪ System Boot ▪ Computer-System Architecture	**SILBER CH 1 STALLINGS CH 1**		**CLO1**
	Operating System Concepts ▪ Multiprogramming, ▪ Multitasking ▪ Multiprocessing ▪ **Interrupts – hardware, software, and exception** o Interrupt Service Mechnism ▪ **Protection** o Memory Protection o IO Protection o CPU Protection ▪ **Dual-mode of Operation:** o User mode o Kernel mode	**SILBER CH** 1 **STALLINGS CH 1**		
2	**Operating system services** o User Interface ▪ CLI ▪ GUI o Programmer Interface ▪ API ▪ System call	**SILBER CH** 2 **STALLINGS CH 2**		
	Operating-System Design ▪ Operating-System structure o Simple o Monolithic o Microkernels o Layered o Modules o Hybrid Systems (Mac OS X, iOS, Android)	**SILBER CH** 2 **STALLINGS CH**		
3	**Processes** ▪ Process ▪ Process States (New, Running, Waiting, Ready and Terminated) ▪ Process Control Block ▪ Process scheduling	**SILBER CH 3 STALLINGS CH 3**	Quiz#1 from Lecture# 1, 2, 3,4	

	○ Scheduling Queues ○ Scheduler (long-term, medium-term and short-term) ▪ Context switching			
4	**UNIX Programming** ▪ Child Process Creation ○ Fork() ▪ Process Invocation ○ Exec() ▪ Orhan & Zombie Processes	**SILBER CH 3** **STALLINGS CH 3**	**Assignment#1**	**CLO2**
5,6	▪ Interprocess communication (IPC) ▪ IPC Methods ○ Anonymous Pipes ○ Shared Memory ▪ **IPC Examples**	**SILBER CH 3** **STALLINGS CH 3**	Quiz#2 from Lecture# 5,6	
7	**CPU Scheduling** • Basic Concepts ○ CPU I/O Burst Cycle ○ Preemptive & Non-preemptive Scheduling • Scheduling Criteria ○ CPU utilization ○ Throughput ○ Turnaround time ○ Waiting time ○ Response time **Scheduling Algorithms** ○ First-Come, First-Served ○ Shortest-Job-First ○ Priority Scheduling	**SILBER CH** 6 **STALLINGS CH 9**		**CLO3**
	Scheduling Algorithms ○ Round-Robin Scheduling ○ Multilevel Queue Scheduling ○ Multilevel Feedback Queue Scheduling	**SILBER CH** 6 **STALLINGS CH 9**		
8	○ **Revision**			
MID-Term Exam				
9	○ **Threads** ▪ What is a Thread? ▪ Single-threaded processes vs. multithreaded processes ▪ Parallelism (Data and Task) ▪ Multithreading Models ○ Many-to-One Model ○ One-to-One Model ○ Many-to-Many Model	**SILBER CH 3,4** **STALLINGS CH 4**		**CLO2**
	UNIX Programming Thread Libraries ○ Pthreads creation	**SILBER CH 4** **STALLINGS CH 4**	**Assignment#2**	

	o Parameter passing to threads o Returing value from threads **Concepts** ▪ Threading Issues o Thread Cancellation ▪ Implicit Threading ▪ Thread Pools			
10	**Process Synchronization** ▪ **The Critical Section Problem** ▪ Solution to the Critical Section Problem o Software Solution ▪ Peterson's Solution o Hardware Solution ▪ Disabling Interrupts ▪ Atomic Instructions o Hybrid ▪ Mutex ▪ Semaphore	**SILBER CH 5** **STALLINGS CH 5**	Quiz#3 from Lecture# 12,13,14	
11	▪ **Mutex** ▪ Producer-Consumer Problem ▪ **Semaphores** o Bounded-Buffer Problem o Reader-Writer Problem o Dining-Philosophers Problem	**SILBER CH 5** **STALLINGS CH 5**		
	Deadlocks ▪ Basic concepts ▪ Deadlock Characterization Deadlock Handling Technique • Deadlock Prevention • Deadlock Avoidance (Banker's Algorithm)	**SILBER CH 7** **STALLINGS CH 6**		
12	**Memory Management** Contiguous memory allocation ▪ Segmentation o Hardware Support o Memory Allocation o Address Translation o Protection o Fragmentation	**SILBER CH 8** **STALLINGS CH 7**	**Assignment#3**	**CLO3**
	▪ Paging o Hardware Support o Memory Allocation o Address Translation	**SILBER CH 8** **STALLINGS CH 7**		

	○ Protection ○ Fragmentaion			
13	▪ Paged-Segmentation ○ Hardware Support ○ Memory Allocation ○ Address Translation ○ Protection ○ Fragmentaion	**STALLINGS CH 7**	Quiz#4 from Lecture# 15-20 (Extra)	
14	**Virtual Memory** ▪ Basic concepts ▪ Demand paging ▪ Performance of demand paging ▪ **Process Creation (copy-on-write and memory-mapped files)**	**STALLINGS CH 8**		
	▪ Page replacement Basic concepts ▪ Replacement algorithms ▪ FIFO ▪ LRU ▪ LFU	**STALLINGS CH 8**		
15	▪ **Revision**			